

A PLAIN-LANGUAGE GUIDE

Modern Epidemiological Study Designs

How epidemiologists compare people — and places — to versions of themselves, to answer questions traditional cohort and case-control studies struggle to reach.

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Based on “Modern Epidemiological Study Designs” by Philip H. Kass, in the Handbook of Epidemiology

ROADMAP

Eight Ideas, One Underlying Trick

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One underlying idea, and the honest limits of every design that uses it

FOUNDATIONS, 2 OF 4

The Big Idea: Comparing a Person to Themselves

Cohort and case-control studies are epidemiology’s workhorses, but they share a basic requirement: finding a comparison group of different people who are similar enough to the people being studied that any difference in outcome can be credited to the exposure, not to all the other ways people differ.

The designs in this chapter take a different route. Instead of asking “how does this exposed person compare to an unexposed person?” they ask “how does this person’s exposed self compare to their own unexposed self, at another time or in another hypothetical version of events?” This is the logic of potential outcomes: imagining the same person, or the same place, under two different conditions.

Comparing someone to themselves automatically cancels out anything about that person that does not change — genetics, personality, lifelong habits, neighborhood history — without ever having to measure or model it. That is the appeal behind case-crossover studies, case-time-control studies, self-controlled case series, and case-specular studies, the four newer “case-only” designs this chapter introduces.

Every Case-Control Study Hides a Cohort

A cohort study starts with exposure and follows people forward to see who develops disease. A case-control study starts with disease and looks backward for exposure. Early critics treated the second as simply the first one reversed — a mix-up memorable enough to earn a name: the “trohoc” fallacy, “trohoc” being “cohort” spelled backward.

The correction: think “cohort-centric”

A well-designed case-control study is not the opposite of a cohort study — it is a cohort study in disguise. Its cases and controls should both be understood as having come from some real, at least conceptually enumerable, underlying cohort: the controls represent the exposure distribution of the source population that produced the cases.

This “cohort-centric” way of thinking is what makes it possible to design smarter, more efficient variations on the case-control study — including several covered later in this chapter — without losing the validity that comes from sampling controls correctly from that hidden cohort.

Self-Comparison Is Not Always the Right Tool

Comparing a person to their own other self only works cleanly when the exposure’s effect is short-lived and the timing of exposure can be pinned down. Two conditions make it unreliable:

A non-intermittent, non-transient exposure

If an exposure is constant — a lifelong trait, a permanent occupational setting — there is no “unexposed self” at another time to compare against.

A long, uncertain induction time

When an exposure’s effect can linger or carry over for a long and poorly known period, it becomes impossible to say confidently which of a person’s many past exposure windows actually caused the event.

In those settings, traditional case-control studies and their cohort-centric variants — covered in the next two sections — remain the right tool. The rest of this chapter is about the settings where self-comparison does work.

CASE-COHORT STUDIES, 1 OF 6

Sample the Cohort Once, Before Anyone Gets Sick

Following an entire large cohort in detail is expensive, especially when exposure requires costly laboratory tests or record review. A case-cohort study economizes by drawing a random “sub-cohort” from the full cohort at the very start — before any cases have occurred — and measuring exposure only in that sub-cohort plus everyone who later becomes a case.

The key design choice: the sub-cohort is picked at time zero and is **not** matched to each case’s failure time. That single difference is what separates a case-cohort study from the nested case-control study covered later in this chapter.

The basic version fixes the sub-cohort at time zero, but the same principle extends to dynamic cohorts where people enter at different times, with the sub-cohort replenished over time as its members are censored.

CASE-COHORT STUDIES, 2 OF 6

Two Practical Advantages

Cheaper exposure ascertainment

Only the sub-cohort and the cases need their exposure measured — not the entire cohort — which matters most when exposure assessment is expensive: gene sequencing, biobank samples, detailed chart abstraction.

One comparison group, many outcomes

Because the sub-cohort was not picked with any single disease in mind, the same sub-cohort can serve as the comparison group for cases of heart disease, cases of cancer, and cases of any other outcome under study, all at once.

That second advantage has no equivalent in a nested case-control study, where a fresh set of controls is normally drawn separately for each outcome of interest.

CASE-COHORT STUDIES, 3 OF 6

Three Ways to Build a Comparison Group

Approach	Controls Drawn	Needs a “Rare Disease” Assumption?	Origin
Cumulative incidence sampling	Non-cases at the end of follow-up	Yes, to approximate the risk ratio	Traditional case-control design (Cornfield, 1951)
Incidence density sampling	Fresh controls at each case’s failure time	No, for the incidence rate ratio	Nested case-control design (Miettinen, 1976)
Case-cohort sampling	One sub-cohort, fixed at time zero	No, and reusable across outcomes	Case-cohort design (Kupper et al., 1975; Miettinen, 1982; Prentice, 1986)

Each row solves the same basic problem — picking a manageable comparison group out of a large cohort — with a different rule for when controls are chosen and how reusable they are.

CASE-COHORT STUDIES, 4 OF 6

Why the Raw Numbers Need Correcting

Because cases are deliberately over-represented relative to the small random sub-cohort, simply comparing raw counts of exposed and unexposed people produces a biased answer. A statistical correction — developed by Sato (1992) — reweights the comparison to account for exactly how imbalanced the sampling was.

1.8

The crude, uncorrected incidence proportion ratio

2.2

The corrected estimate, using Sato's maximum-likelihood method

In this real example (adapted from Miettinen, 1982), the gap between 1.8 and 2.2 is entirely explained by the imbalance between how many cases and how many sub-cohort members were sampled — not by any flaw in the underlying data. The same statistical logic extends to confidence intervals and to combining results across strata.

CASE-COHORT STUDIES, 5 OF 6

Weighting: Making Each Sampled Person Stand In for More

A useful way to picture a case-cohort study, following Breslow and colleagues (2009), is as a two-stage sample: stage one draws the full cohort as a representative sample of the population; stage two decides which of those people get their exposure measured. Because only a fraction of non-cases are sampled into stage two, each sampled non-case is statistically “weighted up” to represent the other similar people who were not sampled — the same logic used to correct an oversampled group in an ordinary survey.

This weighting approach was demonstrated using data from the Atherosclerosis Risk in Communities (ARIC) study, sampled across eight strata defined by ethnicity, sex, and age. Later refinements — calibration, and weights estimated from a logistic regression — improve the statistical precision of the weighted analysis still further.

Because cohort members can join or leave the sub-cohort’s risk set dynamically over time, a case-cohort analysis of the incidence rate ratio needs specialized statistical software rather than an ordinary regression — a theme that recurs throughout this chapter’s designs.

CASE-COHORT STUDIES, 6 OF 6

Case-Cohort or Nested Case-Control?

Choose case-cohort when...

Several different outcomes will be studied from the same cohort, and the same comparison group should serve all of them.

Choose nested case-control when...

There is a single outcome of primary interest, and the study wants an incidence rate ratio estimated with the least fuss from the study's statistical design.

As Langholz and Thomas (1990) and Wacholder (1991) both concluded, the right choice depends less on which design is generally “better” and more on the specific effect measure needed and whether the study has one outcome in mind or several.

NESTED CASE-CONTROL REFINEMENTS, 1 OF 5

A Quick Recap: Risk-Set Sampling

In a nested case-control study, every time a case occurs, one or more controls are drawn from among the cohort members who are still at risk of becoming a case at that exact moment — the case’s “risk set.” The word “nested” signals that the cohort is well-defined and enumerable, so this risk set can actually be identified.

A subtle but important point: being picked as a control does not retire someone from the cohort. They remain eligible to be sampled again as a control for a later case — or, eventually, to become a case themselves.

How many controls to draw per case is itself a design choice: a traditional rule of thumb says little precision is gained beyond four, but that is not universal — the right number depends on exposure prevalence, effect size, and how much matching is used, and some studies gain real value from recruiting more.

NESTED CASE-CONTROL REFINEMENTS, 2 OF 5

Why This Sampling Rule Mattered So Much

Cornfield (1951) showed that an odds ratio from a traditional case-control study approximates the risk ratio — but only if the disease is rare enough in the source population for that approximation to hold. For a common disease, the approximation breaks down.

Miettinen (1976) showed something more powerful: if controls are instead sampled from the risk set at each case's exact failure time — incidence density, or risk-set, sampling — the resulting odds ratio is a valid estimate of the incidence **rate** ratio, with no rare-disease assumption required at all, however common the disease.

That single insight freed epidemiologists from a restrictive assumption and is the statistical foundation the nested case-control design still rests on today.

NESTED CASE-CONTROL REFINEMENTS, 3 OF 5

Counter-Matching: Deliberately Picking Different Controls

Ordinary matching pairs a case with controls who are similar. Counter-matching (Langholz and Borgan, 1995) does the opposite on purpose: it uses an inexpensive, easy-to-measure stand-in for exposure to select controls who are likely to **differ** from the case, guaranteeing more informative, discordant pairs and boosting statistical power without adding subjects.

Mining duration as a stand-in for silica

Steenland and Deddens (1997) found that three counter-matched controls could match the statistical efficiency of ten randomly sampled controls.

Radon and smoking among uranium miners

The Beaverlodge cohort study used counter-matching to sharpen estimates of radon exposure while accounting for smoking.

The gain is largest for rare exposures and disappears — or reverses — if the stand-in is counter-matched on a confounder rather than the exposure itself, unless that stand-in is very strongly related to the true exposure.

NESTED CASE-CONTROL REFINEMENTS, 4 OF 5

Getting More From the Controls You Already Have

Suissa and colleagues (2010) proposed sampling each control's exposure at several different points in time — several “person-moments” — rather than just once. This works for exposures that are acute or transient, though not for ones that build up cumulatively.

The efficiency gain resembles what happens when a study simply recruits more controls: in one comparison, sampling five time windows from two controls delivered roughly the same statistical precision as recruiting four controls with a single time window each — without the added cost of finding new subjects.

NESTED CASE-CONTROL REFINEMENTS, 5 OF 5

Two-Stage Sampling — and Why a Naive Analysis Can Mislead

Two-stage sampling (Breslow and Chatterjee, 1999) measures a cheap variable on everyone, then measures an expensive or difficult one — like a detailed pathology review — only on a deliberately chosen sub-sample.

In the National Wilms' Tumor Study, the entire cohort gave an odds ratio of 4.8. A naive, small 1:1 case-control sample gave a biased 5.8. A weighted sample deliberately including every unfavorable-histology case, analyzed with an ordinary (unweighted) method, gave a badly wrong 0.47. Only a properly weighted regression analysis recovered an estimate close to 4.8 — while checking detailed pathology on just 16% of controls.

The lesson generalizes across every efficient design in this section: sampling cleverly to save money is only half the job. It also requires the matching, specialized statistical analysis — ordinary methods applied to cleverly sampled data can produce a confidently wrong answer.

CASE-CROSSOVER STUDIES, 1 OF 9

The Person Is Their Own Comparison Group

In an experimental crossover trial, a researcher gives the same patient a treatment during one period and a placebo during another, then compares the two. The case-crossover design (Maclure, 1991) is the observational analog: instead of the researcher assigning exposure, the participant’s own life determines when they were exposed and when they were not.

Every subject is a case — someone who has already experienced the outcome. Their exposure right before the event is compared to their own exposure at some other “referent” time. No separate control group of other people is needed at all.

CASE-CROSSOVER STUDIES, 2 OF 9

The “Effect Period”: How Long Could This Exposure Matter?

Every case-crossover study has to define a window of time before the event during which the exposure could plausibly have caused it — the effect period. Biology sets this window: a drug’s half-life, how long a hormone stays active, a vaccine’s window of immune activation, or a pathogen’s incubation time.

Maclure (1991) showed that guessing the effect period wrong — too short or too long — misclassifies exposure and dilutes the true association toward the null. The best-chosen effect period is the one that produces the strongest, clearest effect estimate, since that is the window least distorted by misclassification.

Because the true biological window is rarely known for certain, the chapter recommends testing a range of plausible effect-period lengths as a sensitivity analysis, rather than simply reporting whichever single window happens to produce the largest effect estimate.

CASE-CROSSOVER STUDIES, 3 OF 9

Best Suited to Sudden Events With Episodic Triggers

Good fit

An episodic, short-acting exposure (heavy exertion, a single dose of a drug, a burst of air pollution) paired with an acute, sudden outcome (heart attack, seizure, injury) that occurs uncommonly.

Poor fit

A constant or slowly changing exposure, or an outcome that develops gradually over years — there, an ordinary cohort or case-control study remains the appropriate design.

Within this good-fit territory, the case-crossover design is efficient and intuitive precisely because it needs no separate group of other people to serve as controls.

CASE-CROSSOVER STUDIES, 4 OF 9

Choosing a “Referent” Time, Not a Control Person

The logic borrows directly from a pair-matched case-control study, except every “control” observation comes from the case’s own life. A referent time’s exposure stands in for what the case’s exposure would ordinarily look like, and is compared to what it actually was right before the event.

Using more than one referent time per case increases the number of informative, discordant comparisons — the same statistical benefit that comes from adding more controls in an ordinary case-control study. Researchers have also compared different statistical strategies for handling a recurring pattern, like time of day, when picking referent periods — simpler models are easier to run, while more detailed ones better capture real underlying patterns.

A Worked Example: Exercise and Stroke Risk

Adapted from Maclure (1991): for each person, their usual exercise frequency implies how much of their time is normally spent in a hypothesized two-hour “effect period” right after exercising. That expected, usual exposure share is compared to what actually happened in the two hours before their stroke.

24.0

Estimated incidence rate ratio for a stroke within two hours of aerobic exercise

3.1–188.1

The 95% confidence interval — very wide, reflecting a rare exposure window and a small sample

This approach, called the “usual frequency approach,” assumes there is no trend in the person’s exposure over the retrospective period studied, and that exercise’s effect on stroke risk does not itself vary by time of day.

CASE-CROSSOVER STUDIES, 6 OF 9

The Assumption the Whole Comparison Rests On

Comparing a case's event-time exposure to their referent-time exposure only gives a fair answer if there is no meaningful trend in that person's exposure over time — a condition the chapter calls stationarity. But stationarity alone is not quite the same as the fuller idea of exchangeability, and multiple, ordered referent times raise the bar higher than in an ordinary case-control study, where controls can be freely reordered with no effect on the result.

Stationarity is not the same as exchangeability. Pairwise exchangeability — the event period matched fairly against each referent period, one at a time — is all a Mantel-Haenszel analysis needs, and stationary exposure is usually enough to deliver it. Global exchangeability is stronger: every referent period behaving as interchangeable with every other, not just with the event period. A joint analysis of several referent periods, such as conditional logistic regression, needs that stronger condition.

A concrete failure case: in anti-inflammatory drug use, people self-medicate more during flare-ups of pain and less during periods of relief. Because a person's exposure in one time interval predicts their exposure in the next, global exchangeability breaks down — and a conditional-logistic-regression analysis of the data can suggest a causal effect that is not really there, an effect Vines and Farrington (2001) call “overlap bias.”

CASE-CROSSOVER STUDIES, 7 OF 9

A Fix: Look Both Before and After the Event

Navidi (1998) proposed sampling referent times from both before and after the event, not only before it — a bidirectional design. Averaging across both directions cancels out a gradual trend in exposure that a one-directional (before-only) design would mistake for a real effect.

This requires assuming the event itself does not change the person's future exposure — a reasonable assumption for an exogenous exposure like air pollution, but a shakier one for a health-driven behavior that people might change only because the event just happened to them.

Two hypothetical cohort examples by Navidi (1998) showed that a unidirectional design can produce answers roughly twice as large as the truth when exposure or baseline risk trends over time; a bidirectional design corrected both.

CASE-CROSSOVER STUDIES, 8 OF 9

Four Ways to Pick Referent Times

Each strategy (Bateson and Schwartz, 1999; Lumley and Levy, 2000; Navidi and Weinhandl, 2002) answers the same question — which other moments in a person’s life fairly represent their usual exposure? — with a different trade-off between simplicity and protection against temporal confounding.

Strategy	Referent Times Used	Best For
Full-stratum bidirectional	Every point across the whole ordered observation period	The simplest baseline, assuming one constant rate throughout
Symmetric bidirectional	Two points, equally spaced before and after the event	Day-of-week or similar short-term confounding, controlled by design
Time-stratified	Points drawn from within the same calendar stratum, e.g. the same month	Longer-term trends a symmetric design would still miss
Semi-symmetric bidirectional	One point, chosen at random from the two symmetric options	A simpler single-referent alternative, when temporal effects are constant

A further adjustment (Janes et al., 2005a) — the “adjusted” semi-symmetric bidirectional design — modifies that last approach for events too close to either end of the observation period, where its usual referent time would otherwise fall outside the study altogether.

CASE-CROSSOVER STUDIES, 9 OF 9

One Honest Caution

In a review of 19 air-pollution case-crossover studies, most relied on just two of the four referent-time strategies on the previous slide — but not every methodologist is convinced any of them is the right tool for that setting.

Whitaker and colleagues (2007), reviewing case-crossover methods for air pollution, concluded bluntly that these methods “are either biased or are special cases of more versatile methods” and recommended plain time-series regression instead — a reminder that even a well-established design can face serious, ongoing methodological debate.

CASE-TIME-CONTROL STUDIES, 1 OF 5

When Exposure Itself Is Trending Over Time

Case-crossover studies assume exposure is roughly stationary over the period studied. That assumption fails when exposure has a genuine time trend — most classically in “confounding by indication”: as an illness gets worse, both the intensity of drug treatment and the risk of a bad outcome tend to rise together, even when the drug itself is not to blame.

Disease severity is exactly the kind of confounder that is both hard to measure and a continuum rather than a category, so it usually cannot simply be adjusted for. Suissa (1995) designed the case-time-control study specifically to work around it.

CASE-TIME-CONTROL STUDIES, 2 OF 5

Borrowing a Time Trend From People Who Never Became Cases

Suissa's solution runs a second, parallel case-crossover-style comparison inside a group of controls — people sampled the conventional case-control way, who did not experience the outcome. Their own exposure change between an earlier period and the index period reveals how much of any trend is simply the passage of time, unrelated to disease severity.

That control-group time trend is then used to adjust the case-only estimate, subtracting out the part of the change in exposure that period effects — not the outcome — would have produced anyway. What remains is a cleaner estimate of the exposure's own effect on the outcome.

CASE-TIME-CONTROL STUDIES, 3 OF 5

A Targeted Fix, Not a Cure-All

Like any case-control design, the case-time-control study can still be biased if the controls' exposure distribution does not really represent what the cases' exposure would have looked like without a causal effect. Greenland (1996) laid out further conditions the design needs: the exposure's effect should be stable across time periods, and there should be no unmeasured confounder that itself interacts with time or with exposure.

Those unmeasured-confounder conditions are the hardest to verify in practice, though they tend to be more plausible over shorter rather than longer time periods. The case-time-control study should be understood as a real improvement on a specific, common problem — not a guarantee against every source of confounding.

CASE-TIME-CONTROL STUDIES, 4 OF 5

Where the Design Has Been Put to Work

Hospital safety

Whether serious adverse events in subacute care raise the chance of interhospital transfer

Air pollution

Fine particulate matter (PM2.5) and same-day outpatient visits for hypertension

Pregnancy and COVID-19

The timing and severity of infection during pregnancy and the risk of preterm birth

Drug safety broadly

A recurring tool across pharmacoepidemiology wherever confounding by indication is a concern

Two Further Refinements

Active comparators

Instead of comparing drug users to non-users, compare users of the drug under study to users of an alternative real treatment for the same condition — much closer to how a randomized trial compares two active treatment arms, and a further defense against confounding by indication.

The “case-case-time-control” design

Wang and colleagues (2011) proposed using future cases — people who have not yet had the event but will later — as the comparison group instead of people who never become cases, on the reasoning that future patients may better capture the exposure trend cases themselves experienced.

SELF-CONTROLLED CASE SERIES, 1 OF 6

Built for Studying Rare Vaccine Reactions

Vaccine side effects that are both rare and acute pose a real study-design problem. A cohort study needs a prohibitively large population to catch enough events. A hospital-based case-control study struggles to define a clean source population and a valid control series. Farrington (1995) proposed the self-controlled case series to sidestep both problems.

The design needs only cases — people who experienced the outcome at least once — and no separate comparison group of other people at all.

SELF-CONTROLLED CASE SERIES, 2 OF 6

Comparing a Person's Own Time Windows

Each case's total observed time is divided into a hypothesized "risk period" — the window after exposure when a reaction could plausibly occur, chosen from medical knowledge — and a "control period," covering all of that same person's remaining observed time.

The rate of events during the risk period is compared to the rate during the control period, within the same person. Once again, the comparison is a person against themselves over time, not against anyone else.

SELF-CONTROLLED CASE SERIES, 3 OF 6

Two Reasons This Design Performs So Well

Automatic control for fixed traits

Anything that does not change across a person's observation period — genetic propensity, lifelong environmental exposures, personality — cancels out automatically, without ever needing to be measured.

Nearly as efficient as a full cohort

Farrington (2004) showed that when exposure is not rare, this cases-only design is nearly as statistically efficient as a full study of the entire underlying cohort — despite using only a fraction of the population.

SELF-CONTROLLED CASE SERIES, 4 OF 6

Three Assumptions That Must Hold

The event cannot change future exposure

Having the outcome once should not affect the probability of being exposed again later — if it does, the risk-versus-control comparison is thrown off.

Observation time must be independent of the event

How long a person is followed should not itself be influenced by whether, or when, they experience the outcome.

Case selection cannot depend on exposure

Cases should come from a census of everyone eligible, or have their exposure history obtained only after enrollment — never selected in a way that already knows, and is swayed by, exposure status.

Where any of the three is shaky — for example, if an event could plausibly end future vaccination — researchers have developed adapted versions of the method to compensate.

SELF-CONTROLLED CASE SERIES, 5 OF 6

A Worked Example: Vaccines and Hemolytic Anemia

Picture children followed from age one to three, each vaccinated at some point. Each child's two years of observation is split into a baseline period and three consecutive two-week windows right after vaccination, when an immune-triggered reaction is considered biologically plausible.

Across many children, the count of hemolytic anemia events falling inside those risk windows is compared to the count falling outside them, adjusted for how much observed time each window represents and for the child's age. That comparison is what produces the estimated relative incidence — no separate group of unvaccinated children is required.

If an immediate reaction is not biologically plausible, a short “lag” period with no assumed risk can be inserted between vaccination and the first risk window.

SELF-CONTROLLED CASE SERIES, 6 OF 6

A Design That Adapts to Complex Realities

Multiple risk periods

A multi-component vaccine can have several risk windows evaluated at once

Age and seasonal effects

Underlying risk is allowed to vary by age group, or by season, across the observation period

Recurring events

A person can contribute more than one event, and clustered events can be handled too

Beyond vaccines

The same logic now supports post-marketing safety surveillance for other drugs, with short or long risk periods

CASE-SPECULAR STUDIES, 1 OF 6

When Exposure Is Baked Into Geography

Studying electromagnetic fields from power lines and childhood cancer runs into a stubborn problem: how close a home sits to power lines is not just a measure of field exposure. It is also a proxy for traffic, air pollution, and neighborhood socioeconomic status — all of which can independently affect health. An ordinary neighborhood control group cannot cleanly separate the field's own effect from these other neighborhood effects.

The case-specular design solves this by comparing each case's real home to a hypothetical version of the same home — not to a different, real house down the street.

CASE-SPECULAR STUDIES, 2 OF 6

Building a Home's Own Mirror Image

For each real case residence, researchers construct a “specular” residence: an imaginary house created by reflecting the real one across the center line of the same street. It shares the same neighborhood, the same street, the same everything — except a different, hypothetical distance to the power lines.

Each real home's wire-code exposure category is then compared to its own mirror image's hypothetical wire-code category, analyzed as a matched pair — one real, one purely imaginary.

CASE-SPECULAR STUDIES, 3 OF 6

The Prediction the Design Puts to the Test

If the electromagnetic field hypothesis is correct, real case residences should, on average, carry higher-exposure wire codes than their own mirror-image counterparts — since real homes were not placed with any regard for their imaginary reflection.

If the field hypothesis is wrong, and the earlier apparent associations were really driven by other neighborhood factors, real homes and their mirror images should look statistically the same — because those other factors are shared equally by both.

CASE-SPECULAR STUDIES, 4 OF 6

Three Assumptions Behind a Causal Reading

Symmetry

No systematic reason homes end up closer to power lines on one side of a street than the other

Balanced confounders

Unmeasured neighborhood factors are effectively balanced between the two sides of the street, much like randomization in an experiment

Unbiased classification

Wire codes are assigned to real and specular residences with equal care, with no systematic difference in scrutiny

Zaffanella and colleagues (1998) formalized these conditions, particularly the first — “symmetry of the residence-specular probability matrix” — as necessary for the comparison to support a causal conclusion.

CASE-SPECULAR STUDIES, 5 OF 6

What This Buys — and What It Costs

Strengths

No real control group needs to be recruited or measured, which cuts cost and avoids an entirely separate source of selection bias.

Limits

Constructing a specular residence can be contrived on unusually shaped streets; when power lines run behind homes, real and specular wire codes often turn out identical and uninformative; and the crucial symmetry assumption cannot itself be verified without a real validation control group.

CASE-SPECULAR STUDIES, 6 OF 6

The Same Trick, Applied to a New Question

The “mirror” logic extends naturally beyond residences. Larjavaara and colleagues (2011) studied mobile phone use and the location of gliomas — a type of brain tumor — by comparing each tumor’s actual location to a hypothetical location obtained by reflecting it across the brain’s coronal and axial planes.

This let researchers test whether tumors cluster nearer to where a phone is typically held, using the same within-subject, real-versus-hypothetical comparison at the heart of the original power-line studies.

BRINGING IT TOGETHER, 1 OF 4

Two Branches of the Same Modern Logic

The chapter's designs split into two families, not one. Case-cohort studies and the nested case-control refinements — counter-matching, multiple time windows, two-stage sampling — still compare different people sampled from a shared underlying cohort; what makes them modern is how cleverly and efficiently that sampling and analysis is done.

The other family — case-crossover, case-time-control, self-controlled case series, and case-specular studies — goes further: it replaces that external comparison group with a within-person or hypothetical self-comparison instead. As the foundations section noted, that swap only suits short-acting, transient exposures; where it does not fit, the chapter deliberately falls back on the cohort-sampling family.

BRINGING IT TOGETHER, 2 OF 4

A Quick Guide to Which Design Fits

If the situation is...	Consider...
Several outcomes to study from one cohort, expensive exposure data	A case-cohort study
One outcome, and squeezing the most precision from a nested case-control study	Counter-matching, multiple time windows, or two-stage sampling
A short-acting exposure and an acute, sudden outcome	A case-crossover study
The same, but exposure has a meaningful trend over time	A case-time-control study
A very rare, acute event with a suspected transient trigger, like a vaccine	A self-controlled case series
Exposure that is geographically determined and confounded with neighborhood factors	A case-specular study

BRINGING IT TOGETHER, 3 OF 4

New Assumptions, Not Fewer Assumptions

For the designs that trade an external comparison group for a within-person or hypothetical self-comparison, that trade does not make assumptions disappear — it exchanges one set for another: stationarity of exposure, exchangeability across time periods, no unmeasured confounder that itself interacts with time, and symmetry in how a hypothetical comparison was constructed. The cohort-sampling family carries its own version of the same lesson: sampling controls more cleverly is not free either, and demands matching, specialized statistical analysis to avoid bias, as the Wilms’ Tumor example showed.

These new assumptions are not automatically safer than the old ones. As the chapter’s own internal debate over case-crossover methods for air pollution shows — where respected methodologists have recommended against the design in favor of simpler time-series regression — each assumption has to be checked against the specific study at hand, not simply presumed because the design sounds clever.

A practical response, drawn from the case-time-control literature: when it is unclear which design’s assumptions are least likely to be violated, analyze the same data with two or more of these designs as a sensitivity analysis, and treat agreement — or disagreement — across their different assumptions as evidence in its own right.

BRINGING IT TOGETHER, 4 OF 4

Compare a person, or a place, to itself, and a whole set of confounders disappears automatically — but a new set of assumptions takes its place, and those have to be checked just as carefully.

As epidemiologists put these newer designs to work alongside cohort and case-control studies, the resulting conflicts in findings are not a sign of failure — they are the ordinary, healthy friction of a field getting better at asking causal questions.

SOURCE

Kass, P. H. “Modern Epidemiological Study Designs.” In the Handbook of Epidemiology. Springer Science+Business Media, LLC, part of Springer Nature.

This deck condenses the chapter’s case-cohort, nested case-control, case-crossover, case-time-control, self-controlled case series, and case-specular study designs into plain language. Every named study, statistic, and worked example is drawn directly from the original chapter text.